

0 · THE VOCABULARY

	Population	Sample
Size	N	n
Mean	μ	$\bar{x}$
Stdev	σ	S
Essence	a constant	a random variable
Known?	never	yes, after sampling
Theory. A sample is IID (independent, identically distributed). We never observe μ or σ — we estimate them. That is why the sample variance divides by n–1 : $\bar{x}$ was itself estimated from the same data, costing one degree of freedom.		

1 · CENTRE

$\bar{x} = \Sigma x / n = \text{AVERAGE}()$
Median = middle =MEDIAN()
Odd n → middle value. Even n → average the two middle ones.
Outliers. The mean is highly sensitive; the median resists. Bill Gates enters the bar → mean explodes, median barely moves. 2,5,7,9, 10 → med 7, mean 6.6. 2,5,7,9, 10000 → med 7, mean 2004.6.
Skewed data → quote the median as “typical”.

2 · SPREAD

$S^2 = \Sigma (x - \bar{x})^2 / (n-1) = \text{VAR}()$
 $S = \sqrt{S^2} = \text{STDEV}()$

Why square? Raw deviations from the mean always sum to **zero** — positives cancel negatives. Squaring kills the sign; the root returns it to the original units. Stdev answers “what is the average distance from the mean?”

Excel	÷	Use
STDEV / .S	n-1	course default
STDEVP / .P	n	data IS the population

2,5,7,9,10 → $\Sigma (x - \bar{x})^2 = 41.2$, $S^2 = 10.3$, $S = 3.21$. Same mean, different spread = different risk.

3 · SKEWNESS — lopsided?

$=\text{SKEW}() \approx 3(\text{Mean} - \text{Median}) / \text{Stdev}$		
Skew	Order	Means
+ right	Mode < Med < Mean	a few very LARGE
0	Mean = Med = Mode	symmetric
– left	Mean < Med < Mode	a few very SMALL
 skew ≥ 0.5 is significant.		
Shortcut. The mean is dragged toward the long tail; the median stays. So Mean > Median ⇒ right skew .		

Right: income (mean 106.5k > med 83.7k), box office, market caps. Left: easy grades, satisfaction scores.

4 · KURTOSIS — fat tails?

$=\text{KURT}()$
TRAP. Excel KURT gives EXCESS kurtosis (Normal = 0). The lectures use the scale where Normal = 3. Add 3.
Theory. Kurtosis measures tail weight , not direction (that is skew). High kurtosis = higher peak AND heavier tails than a Normal with the same σ — extremes in both directions are more common than Normality predicts.
Why it matters. Heavy tails mean a Normal approximation understates disaster risk. S&P 500 monthly $\mu = 0.66\%$, $\sigma = 3.5\%$ → March 2020 was a –6σ event.

Agent A kurt 3 = Normal; agent B kurt 5 → more very-long and very-short calls, so Normality is riskier for B.

5 · CORRELATION

$=\text{CORREL}(\text{Xrange}, \text{Yrange}) \quad -1 \leq \text{Corr} \leq 1$	
$\text{Cov} = \text{Corr} \cdot \sigma_x \cdot \sigma_y \quad \quad \text{Corr} = \text{Cov} / (\sigma_x \sigma_y)$	
Theory. Corr is a linear relationship index. Three requirements: bounded [–1,1]; the sign gives direction; the magnitude gives how tightly the points hug a straight line.	
Value	Meaning
+1	perfect positive line
0	no linear relationship
–1	perfect negative line
1. Corr = 0 ≠ independent. A circle, parabola or wave all give Corr = 0 while being perfectly dependent. It only detects straight lines .	
2. Corr ≠ slope. 0.30 does NOT mean “30 cents per dollar”.	
3. Corr ≠ causation. Foot size vs education = 0.849.	
4. Correlations break in crises. LTCM 1998: Russia defaulted, historical relationships inverted, –\$4.6bn in four months.	

T/F drill (Corr 0.30, restaurants vs fitness): positive linear relationship ✓ · below-average on one ⇒ expect below-average on the other ✓ · “30 cents per dollar” ✗ · proves causation ✗

6 · QUIZ ANSWERS

Q. Mean market cap >> median → ?
A. Right skew ; a few giants pull the mean up.
Q. Satisfaction skew –3.2 → ?
A. Most are satisfied , a small number very dissatisfied (long LEFT tail).
Q. Kurtosis 3 vs 5, same mean/stdev → ?
A. B has more extremes both ways; Normal has kurtosis 3, so Normality is riskier for B .

20 · STANDARD NORMAL P(Z < z)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

Negative z: $P(Z < -z) = 1 - P(Z < z)$.

7 · Z-SCORE — the master tool

$$z = (x - \text{mean}) / \text{stdev}$$

Theory. z is the distance from the mean measured in **stdev units**. It rescales any variable onto one common ruler, so outcomes from different distributions become comparable. **Unusualness = |z|**. Works for ANY distribution; only turning z into a percentile needs Normality.

$$= \text{NORMSDIST}(z) \rightarrow \% \text{ BELOW} = \text{NORMSINV}(p) \rightarrow z$$

Shanghai	NYSE
$x=11.75, \mu=0.57, \sigma=7.25$	$x=10.39, \mu=0.46, \sigma=4.49$
$z = 1.54$	$z = 2.21 \leftarrow \text{rarer}$

The **bigger** raw return was the **less** exceptional event. Always compare |z|.

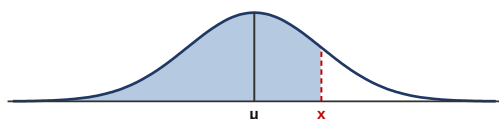
Score 90 in $N(75, 10)$: $z=1.5 \rightarrow 0.9332$ (93rd pct). In $N(80, 10)$: $z=1.0 \rightarrow 0.8413$ (84th).

8 · NORMAL — ALWAYS DRAW IT

bell → mark μ → mark x → **shade**

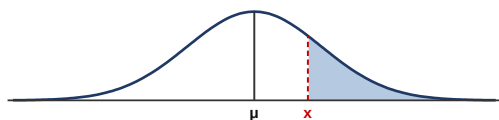
Shaded **LEFT** → NORMDIST straight off. Shaded **RIGHT** → $1 - \text{NORMDIST}$. The picture tells you whether to subtract.

1 · $P(X < x)$ — shade **LEFT**



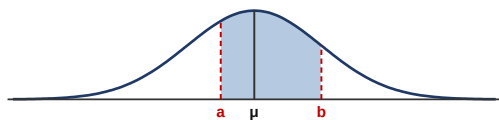
$$= \text{NORMDIST}(x, \mu, \sigma, \text{TRUE})$$

2 · $P(X > x)$ — shade **RIGHT**



$$= 1 - \text{NORMDIST}(x, \mu, \sigma, \text{TRUE})$$

3 · $P(a < X < b)$ — shade **MIDDLE**



$$= \text{NORMDIST}(b, \dots) - \text{NORMDIST}(a, \dots)$$

$X \sim N(80, 10)$: $P(X < 90) = .8413$ · $P(X > 90) = .1587$ · $P(75 < X < 90) = .5328$

9 · BACKWARDS — NORMINV

the **AREA** is known, x is not



$$x = \text{NORMINV}(p, \mu, \sigma)$$

TRAP. The shaded tail is **0.10**, but **NORMINV** wants the area to its **LEFT**: **0.90**.

Asked for	Feed NORMINV
Top 10% (High Pass)	0.90 → 92.8
Top 5%	0.95
Bottom 5% (worst loss)	0.05
Service level 99%	0.99

VaR: $\mu=7\%$, $\sigma=10\%$, worst 5% → $\text{NORMINV}(0.05, 7, 10) = -9.45\%$. Common z :
 90% → 1.28 · 95% → 1.645 · 99% → 2.33

10 · THE 68-95-99.7 RULE

Interval	Inside	Outside
$\mu \pm 1\sigma$	68.3%	31.7%
$\mu \pm 2\sigma$	95.4%	4.6%
$\mu \pm 3\sigma$	99.7%	0.3%

Weight $\mu=180, \sigma=10 \rightarrow 95.4\%$ of men are 160–200 lbs. Probability = **area under the curve**; $P(X = x) = 0$ for a continuous variable.

11 · RANDOM VARIABLES — $E(X)$ and $V(X)$

$$E(X) = \mu = \sum x \cdot P(X=x)$$

$$V(X) = \sigma^2 = \sum P(x) \cdot [x - E(X)]^2$$

$$\sigma = \sqrt{V(X)}$$

$E(X)$ is the probability-weighted average · the long-run average · the fair price · the **balance point** of the distribution.

x	P(x)	x · P	$P(x - E)^2$
0	.1	0	.1(0 - 1.8) ² = .324
1	.3	.3	.3(1 - 1.8) ² = .192
2	.3	.6	.3(2 - 1.8) ² = .012
3	.3	.9	.3(3 - 1.8) ² = .432
Σ	1.0	1.8	0.96

$$E = 1.8 \cdot V = 0.96 \cdot \sigma = 0.9797$$

Square the gap **before** weighting. $V(X)$ is never negative; $V(X)=0$ means X is a constant.

12 · LINEAR TRANSFORM $Y = a + bX$

$$E(a + bX) = a + b \cdot E(X)$$

$$V(a + bX) = b^2 \cdot V(X)$$

$$\text{stdev}(a + bX) = |b| \cdot \text{stdev}(X)$$

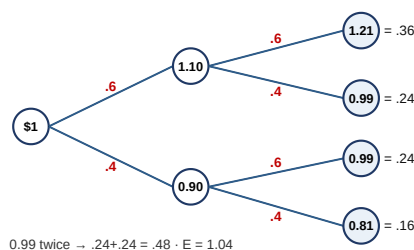
a slides the distribution sideways. It **never** changes the spread.

$Y = -50 + 200X \rightarrow E = -50 + 200(1.8) = 310$; $V = 200^2(0.96) = 38,400$; $\sigma = 200(0.9797) = 195.96$.

A **non-linear** payoff (tiered bonus, a cap) breaks this rule. Rebuild the table: $E(g) = \sum g(x)P(x)$. Bonus 0/15/52.5/82.5 → $E = 45$, $\sigma = 30.187$.

13 · PROBABILITY TREES

ALONG a path × (**AND**) · **BETWEEN** paths + (**OR**)



0.99 twice → .24 + .24 = .48 · $E = 1.04$

1. A branch per outcome; write the probability **on** the branch.
2. Multiply along a path.
3. Add paths that give the same answer.
4. All path probabilities must sum to **1**.

$$P(\text{never}) = (1 - p)^N$$

$$P(\text{at least one}) = 1 - (1 - p)^N$$

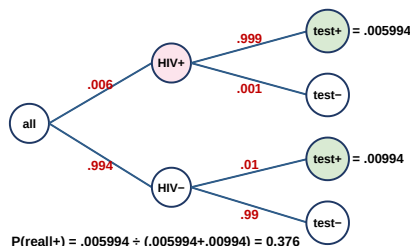
Never add up the “at least one” cases. There is only **ONE** never-path — compute that and subtract. AAA bond $p=.0002$, 25 yrs → **0.004988**.

Two dice: 36 paths, $E(\text{sum})=7$. Chuck-a-luck: $E = -\$0.0463$, $\sigma = 1.288$.

14 · BAYES — reverse probability

$$P(A|+) = P(A) \cdot P(+|A) \div P(+)$$

$$P(+)=P(A)P(+|A)+P(\text{not}A)P(+|\text{not}A)$$



A 99.9% accurate test for a **rare** condition is still mostly **false alarms** — the healthy branch is enormous. Split by **condition first**, then by test result.

15 · SAXBY — TWO VARIABLES

$$E(aX+bY) = a \cdot E(X) + b \cdot E(Y)$$

$$V(aX+bY) = a^2V(X) + b^2V(Y) + 2ab \cdot \sigma_X \cdot \sigma_Y \cdot \text{Corr}$$

$$\text{stdev} = \sqrt{V}$$

Question asks about	a	b
total X+Y	1	1
difference / P(X>Y)	1	-1
portfolio, w in X	w	1-w
weighted 35 / 65	.35	.65

TRAP. stdev is **NEVER** added. $\text{stdev}(aX+bY) \leq a\sigma_X + b\sigma_Y$, equality only at Corr = +1. That gap IS diversification.

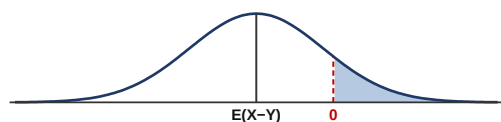
Cross term. Corr > 0 raises risk · Corr = 0 kills the term · Corr < 0 lowers it. With **b negative** the term **subtracts**.

$$N(100,10^2) + N(150,15^2), \text{Corr} .8 \rightarrow V = 100+225+240 = 565, \sigma = 23.77, E = 250.$$

16 · SAXBY + NORMAL = the big one

Step 1 SAXBY → Step 2 draw ONE bell

P(X > Y): mark ZERO on the X-Y curve



$$P(X > Y) = P(X - Y > 0)$$

a=1, b=-1 → E = 100-150 = -50, $\sigma = \sqrt{(100+225-240)} = 9.22$. Zero is 5.4σ above the mean → P ≈ 0.

$$\text{stock} = \text{NORMINV}(0.99, \mu_{\text{total}}, \sigma_{\text{total}})$$

Total demand N(250, 23.77) → 305.3 → order **306**. Inventory answers **round UP**.

Weighted scores: SAI 35% / NAI 65%, both N(100,16²). Ann → 121.6. Corr 1 → σ=16, pct .9115. Corr 0.5 → σ=14.1, pct .937. Lower Corr shrinks σ and **raises** the percentile.

17 · PORTFOLIO P = wX + (1-w)Y

$$E(P) = w\mu_X + (1-w)\mu_Y$$

$$V(P) = w^2\sigma_X^2 + (1-w)^2\sigma_Y^2 + 2w(1-w)\sigma_X\sigma_Y \cdot \text{Corr}$$

① **Target RETURN** (linear, one answer)

$$w = (\text{target} - \mu_Y) / (\mu_X - \mu_Y)$$

10% / 20% assets, target 15% → w = 0.5.

② **Target RISK** (quadratic, TWO answers)

$$A = \sigma_X^2 + \sigma_Y^2 - 2 \cdot \text{Corr} \cdot \sigma_X \sigma_Y$$

$$B = 2 \cdot \text{Corr} \cdot \sigma_X \sigma_Y - 2\sigma_Y^2$$

$$C = \sigma_Y^2 - \text{target}^2 \quad w = (-B \pm \sqrt{B^2 - 4AC}) / 2A$$

σ 15%/35%, Corr .5, target 20% → w = 0.60. Take the root inside [0,1].

③ **MINIMUM-RISK portfolio**

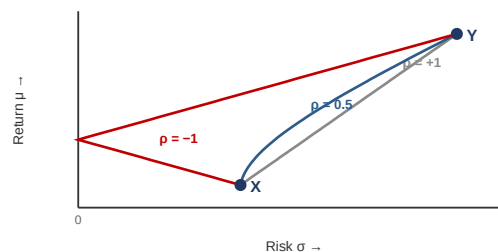
$$w^* = (\sigma_Y^2 - \text{Corr} \cdot \sigma_X \sigma_Y) / (\sigma_X^2 + \sigma_Y^2 - 2 \cdot \text{Corr} \cdot \sigma_X \sigma_Y)$$

σ 3.16 & 2.68, Corr -0.5 → w* = 0.44, min σ = 1.45 — below BOTH assets.

If w* falls outside [0,1] it needs **short selling**; answer with the boundary (0 or 1).

18 · DIVERSIFICATION & POOLING

Lower correlation bends the curve LEFT



Corr	The frontier
+1	straight line — no benefit
0	bends left — real reduction
-1	risk can reach zero

Portfolio effect: a diversified portfolio can have lower volatility than ANY single asset, because idiosyncratic randomness averages out. The **efficient frontier** is the upper-left boundary — best return for each risk level (Markowitz 1952).

Can the portfolio...	
return more than the better asset?	NO
return less than the worse asset?	NO
have stdev BELOW both assets?	YES

Pooling: separate $\Sigma(\mu+k\sigma)$ vs $\Sigma\mu + k \cdot \sigma(X+Y)$

GPUs μ=20, σ=10 each, Corr .2, k=3: separate 50+50 = **100**; pooled σ = √240 = 15.49 → 40+3(15.49) = **86.5**. Saves 13.5. The means still add — it is the **stdev** that shrinks.

19 · DECISION TREE & FINAL CHECKLIST

Question gives you	Do
a list of raw numbers	AVERAGE / STDEV / SKEW
x, μ, σ → “what %?”	z then NORMSDIST
a % → “what score?”	NORMINV
outcomes + probabilities	ΣxP and ΣP(x-E)²
Y = a + bX	a+bE(X), b σ
stages / “at least one”	tree: 1-(1-p)^N
“given the test was +”	Bayes
a correlation	SAXBY
two RVs + a probability	SAXBY then NORMDIST
weights / target return / risk	portfolio formulas

1. STDEV (sample), not STDEVP. **2.** KURT needs +3. **3.** “Greater than” → 1-NORMDIST. **4.** “Top p%” → feed NORMINV 1-p. **5.** For X-Y, b=-1; cross term subtracts. **6.** Did you take the √? Variance ≠ stdev. **7.** μ and σ on the same scale. **8.** Units → round UP. **9.** Two roots → pick the one in [0,1]. **10.** Write the formula in Canvas for partial credit.